

Assignment 4 Question 1

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1 Library Setup

```
library(dplyr)
library(tidyr)
library(lubridate)
library(stringr)
library(ggplot2)
library(tinytex)
```

2 Question 1 (a)

```
#Q1.a

# Load the dataset
propertydataset <- read.csv("PropertyDataset1.csv")

# Check the first few rows of sale_date to confirm its format
head(propertydataset$sale_date)

## [1] "29/05/2020" "17/12/2019" "09/02/2017" "19/12/2018" "31/03/2017"
## [6] "08/03/2019"

# Convert sale_date to Date type (assuming the format is DD/MM/YYYY)
propertydataset$sale_date <- dmy(propertydataset$sale_date)

# Check if the conversion worked by printing the first few rows of the sale_date column
head(propertydataset$sale_date)

## [1] "2020-05-29" "2019-12-17" "2017-02-09" "2018-12-19" "2017-03-31"
## [6] "2019-03-08"

# Check for NA values in sale_date after conversion
sum(is.na(propertydataset$sale_date))

## [1] 0

#Change to numeric
propertydataset$num_bedrooms <- as.numeric(gsub("[^0-9]", "", propertydataset$num_bedrooms))
propertydataset$num_bathrooms <- as.numeric(gsub("[^0-9]", "", propertydataset$num_bathrooms))

# Filter for houses sold in 2017 with 3 bedrooms and 2 bathrooms
house_2017 <- propertydataset %>%
  filter(year(sale_date) == 2017, num_bedrooms == 3, num_bathrooms == 2)

# View the filtered data
head(house_2017)

##           geohash  sale_date  price
## 1 gc7ruev07v23 2017-11-10 270000
## 2              2017-02-17 190000
## 3 gc7rv2wws1w1 2017-12-21 243000
## 4 gc73pms7gejb 2017-06-07 178700
## 5              2017-08-03 305000
## 6              2017-11-29 130000
##                                     formatted_address
## 1      37 The Green, Dunboyne Castle, Dunboyne, Co. Meath
## 2 128 Ashfield East, Johnswell Rd, Kilkenny, Co. Kilkenny
## 3              37 The Crescent, Ongar, Co. Dublin
## 4              33 Westwinds, Kilrane, Co. Wexford
```

```

## 5 6 Woodbrook Park, Riverwood Rd, Castleknock, Co. Dublin
## 6          14 Cloverhill Close, Killorglin, Co. Kerry
##          raw_address post_code   county num_bedrooms
## 1      37 THE GREEN, DUNBOYNE CASTLE, DUNBOYNE           Meath           3
## 2     128 ASHFIELD EAST, JOHNSWELL RD, KILKENNY           Kilkenny          3
## 3     37 THE CRESCENT, ONGAR VILLAGE, DUBLIN 15 Dublin 15     Dublin           3
## 4          33 WESTWINDS, KILRANE, WEXFORD           Wexford           3
## 5 6 WOODBROOK PARK, RIVERWOOD RD, CASTLEKNOCK Dublin 15     Dublin           3
## 6          14 CLOVERHILL CLOSE, KILLORGLIN, KERRY           Kerry           3
## num_bathrooms          property_description property_size_description
## 1          2 Second-Hand Dwelling house /Apartment
## 2          2 Second-Hand Dwelling house /Apartment
## 3          2 Second-Hand Dwelling house /Apartment
## 4          2 Second-Hand Dwelling house /Apartment
## 5          2 Second-Hand Dwelling house /Apartment
## 6          2 Second-Hand Dwelling house /Apartment
## formatted_description not_full_market_price vat_exclusive parking parking1
## 1 Semi-Detached House           FALSE           FALSE     Has Parking
## 2 Semi-Detached House           FALSE           FALSE     Has Parking
## 3 Duplex For Sale                FALSE           FALSE     No Parking
## 4 Bungalow For Sale              FALSE           FALSE     Has Parking
## 5 Duplex For Sale                FALSE           FALSE     Has Parking
## 6 Bungalow For Sale              FALSE           FALSE     Has Parking

```

3 Question 1 (b)

```
#Q1.b

# Extract numeric property size from the description and create a new numeric column
propertydataset <- propertydataset %>%
  mutate(
    property_size_numeric = as.numeric(str_extract(property_size_description, "\\d+"))
  )

# Filter data for premises sold in Cork or Galway and greater than 125 square metres
cork_galway_data <- propertydataset %>%
  filter(
    county %in% c("Cork", "Galway"), # Filter for Cork or Galway
    property_size_numeric == 125      # Area greater than 125 square metres
  ) %>%
  arrange(desc(price))               # Order by price from highest to lowest

# View the filtered and sorted data
head(cork_galway_data)
```

```
##   geohash   sale_date   price formatted_address
## 1          2012-12-06 3378855
## 2          2015-10-20 1192883
## 3          2018-08-01  820000
## 4          2013-07-01  817015
## 5          2017-02-07  750000
## 6          2018-04-13  722467
##                                     raw_address post_code county
## 1 Block L   Castletreasure, Temple Grove, Carr's Hill Douglas      Cork
## 2                                     Clonmore, Mallow      Cork
## 3                                     3 Steeplewoods, Killumney Road, Carrigrohane      Cork
## 4                                     4 Kilconor, Model Farm Road, Cork      Cork
## 5                                     ballinvoher, turloughmore      Galway
## 6                                     19 The Grove, Goatstown Road, Dublin 14      Galway
##   num_bedrooms num_bathrooms   property_description
## 1           NA           NA New Dwelling house /Apartment
## 2           NA           NA New Dwelling house /Apartment
## 3           NA           NA New Dwelling house /Apartment
## 4           NA           NA New Dwelling house /Apartment
## 5           NA           NA New Dwelling house /Apartment
## 6           NA           NA New Dwelling house /Apartment
##               property_size_description formatted_description
## 1               greater than 125 sq metres
## 2               greater than 125 sq metres
## 3 greater than or equal to 125 sq metres
## 4               greater than 125 sq metres
## 5 greater than or equal to 125 sq metres
## 6 greater than or equal to 125 sq metres
##   not_full_market_price vat_exclusive parking parking1 property_size_numeric
## 1                FALSE      FALSE    Has Parking      125
## 2                 TRUE      TRUE    Has Parking      125
```

## 3	FALSE	TRUE	No	Parking	125
## 4	FALSE	TRUE	Has	Parking	125
## 5	FALSE	FALSE	Has	Parking	125
## 6	FALSE	TRUE	Has	Parking	125

4 Question 1 (c)

```
#Q1.c

# Filter for new dwelling houses or apartments in the specified areas
new_dwelling_house_apt <- propertydataset %>%
  filter(
    county == "Dublin", # Ensure the county is Dublin
    grepl("Dublin 18|Dublin 16|Dublin 15|Dublin 8|Dublin 4", post_code),
    grepl("New Dwelling house|New Apartment", property_description)
  ) %>%
  summarise(
    avg_price = mean(price, na.rm = TRUE), # Calculate the average price
    max_price = max(price, na.rm = TRUE), # Calculate the maximum price
    min_price = min(price, na.rm = TRUE) # Calculate the minimum price
  )

# View the resulting summary data frame
print(new_dwelling_house_apt)

##   avg_price max_price min_price
## 1  404612.8  2048458     8000
```

5 Question 1 (d)

#Q1.d

View unique values in the property_description column

```
unique(propertydataset$property_description)
```

```
## [1] "New Dwelling house /Apartment"
## [2] "Second-Hand Dwelling house /Apartment"
## [3] "Teach/Árasán Cónaithe Atháimhe"
## [4] "second-hand dwelling house /apartment"
## [5] "second-hand dwelling house / apartment"
```

View the frequency of each value in the property_description column

```
table(propertydataset$property_description)
```

```
##
##      New Dwelling house /Apartment second-hand dwelling house / apartment
##                                5847                                1
## second-hand dwelling house /apartment Second-Hand Dwelling house /Apartment
##                                2                                34146
##      Teach/Árasán Cónaithe Atháimhe
##                                4
```

View unique values in the property_description column

```
unique(propertydataset$county)
```

```
## [1] "Leitrim"      "Dublin"      "Kilkenny"    "Kildare"     "Tipperary"
## [6] "Louth"       "Limerick"    "Cork"        "Clare"       "Wexford"
## [11] "Roscommon"   "Kerry"       "Waterford"   "Sligo"       "Westmeath"
## [16] "Wicklow"     "Donegal"     "Offaly"      "Mayo"        "Meath"
## [21] "Carlow"      "Longford"    "Cavan"       "Laois"       "Galway"
## [26] "Monaghan"    "westmeath"   "dublin"      "Corcaigh"    "Dubln"
## [31] "cork"        "Ros Comain"  "wicklow"     "kildare"     "Tipp "
## [36] "limerick"    "donegal"
```

View the frequency of each value in the property_description column

```
table(propertydataset$county)
```

```
##
##      Carlow      Cavan      Clare      Corcaigh      cork      Cork      donegal
##          427          540          892          3          1          4176          1
##      Donegal      dublin      Dublin      Dubln      Galway      Kerry      kildare
##          1075          2          13160          1          2034          1176          2
##      Kildare      Kilkenny      Laois      Leitrim      limerick      Limerick      Longford
##          1953          667          561          369          1          1495          327
##      Louth      Mayo      Meath      Monaghan      Offaly      Ros Comain      Roscommon
##          1076          954          1617          322          472          2          570
##      Sligo      Tipp      Tipperary      Waterford      westmeath      Westmeath      Wexford
##          615          2          1087          1040          1          749          1347
##      wicklow      Wicklow
##          1          1282
```



```
# Standardize all variations of "second-hand dwelling house/apartment" to  
"second-hand dwelling house/apartment"
```

```
## [1] "second-hand dwelling house/apartment"
```

```
propertydataset$property_description <- gsub(  
  pattern = "second-hand dwelling house /  
  apartment|second-hand dwelling house /  
  Apartment|Second-Hand Dwelling house /  
  Apartment|second-hand dwelling house /Apartment",  
  replacement = "second-hand dwelling house/apartment",  
  x = propertydataset$property_description,  
  ignore.case = TRUE  
)
```

```
# Check the first few rows to ensure the replacement worked  
head(propertydataset$property_description)
```

```
## [1] "New Dwelling house /Apartment"  
## [2] "second-hand dwelling house/apartment"  
## [3] "second-hand dwelling house/apartment"  
## [4] "second-hand dwelling house/apartment"  
## [5] "second-hand dwelling house/apartment"  
## [6] "New Dwelling house /Apartment"
```

```
# Check to ensure the changes have been applied correctly  
head(propertydataset$property_description)
```

```
## [1] "New Dwelling house /Apartment"  
## [2] "second-hand dwelling house/apartment"  
## [3] "second-hand dwelling house/apartment"  
## [4] "second-hand dwelling house/apartment"  
## [5] "second-hand dwelling house/apartment"  
## [6] "New Dwelling house /Apartment"
```

```
# Create a named vector for standardizing county names  
county_map <- c(  
  "Leitrim" = "Leitrim",  
  "Dublin" = "Dublin",  
  "Kilkenny" = "Kilkenny",  
  "Kildare" = "Kildare",  
  "Tipperary" = "Tipperary",  
  "Louth" = "Louth",  
  "Limerick" = "Limerick",  
  "Cork" = "Cork",  
  "Clare" = "Clare",  
  "Wexford" = "Wexford",  
  "Roscommon" = "Roscommon",  
  "Kerry" = "Kerry",  
  "Waterford" = "Waterford",
```

```

"Sligo" = "Sligo",
"Westmeath" = "Westmeath",
"Wicklow" = "Wicklow",
"Donegal" = "Donegal",
"Offaly" = "Offaly",
"Mayo" = "Mayo",
"Meath" = "Meath",
"Carlow" = "Carlow",
"Longford" = "Longford",
"Cavan" = "Cavan",
"Laois" = "Laois",
"Galway" = "Galway",
"Monaghan" = "Monaghan",
"westmeath" = "Westmeath",
"dublin" = "Dublin",
"Corcaigh" = "Cork",
"Dubln" = "Dublin",
"cork" = "Cork",
"Ros Comain" = "Roscommon",
"wicklow" = "Wicklow",
"kildare" = "Kildare",
"Tipp " = "Tipperary",
"limerick" = "Limerick",
"donegal" = "Donegal"
)

# Use recode to standardize county names
propertydataset$county <- recode(propertydataset$county, !!!county_map)

# Check the first few rows to ensure the replacement worked
head(propertydataset$county)

## [1] "Leitrim"    "Dublin"     "Kilkenny"   "Kildare"    "Tipperary"  "Kildare"

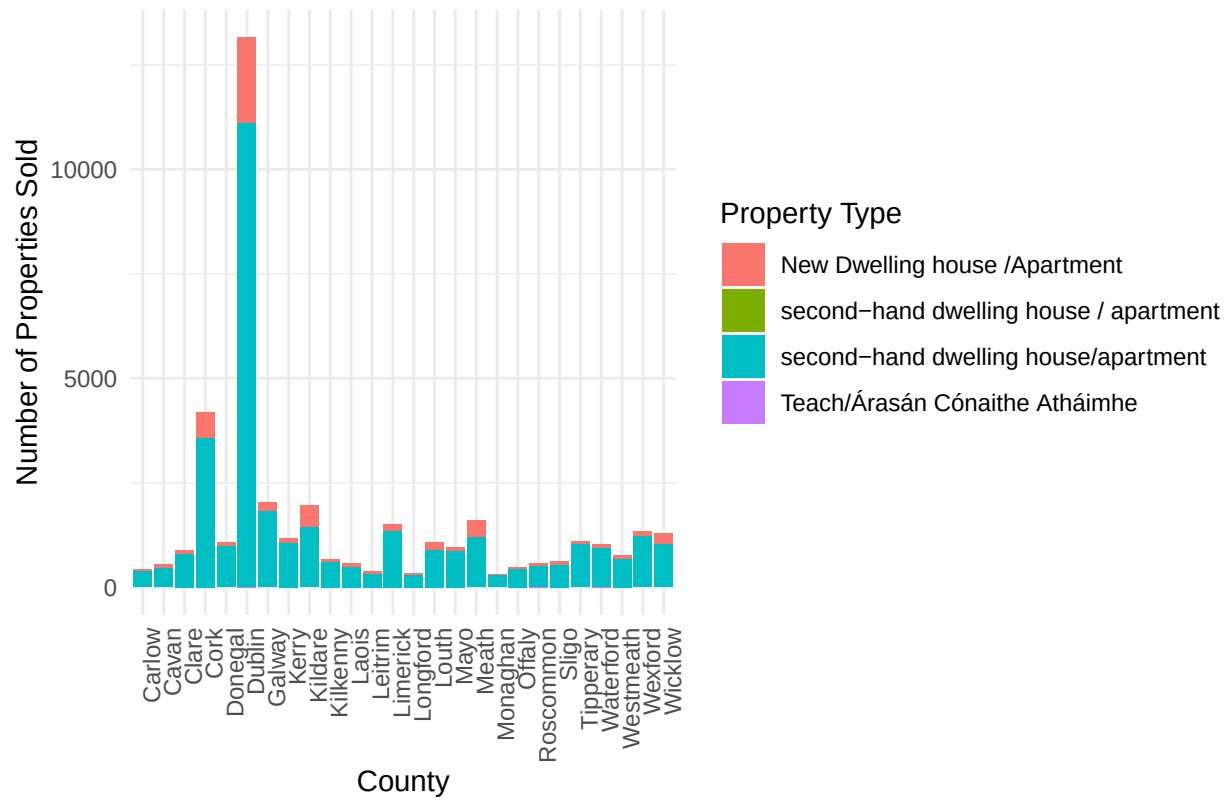
```

```

# Create a bar chart showing the number of properties sold in different counties
# Color the bars based on property type (e.g., house or apartment)
ggplot(propertydataset, aes(x = county, fill = property_description)) +
  geom_bar() + # Use a bar chart
  labs(
    title = "Properties Sold by County in Ireland",
    x = "County",
    y = "Number of Properties Sold",
    fill = "Property Type" # Label for the legend
  ) +
  theme_minimal() + # Use a minimal theme
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1)
  ) # Rotate x-axis labels for readability

```

Properties Sold by County in Ireland

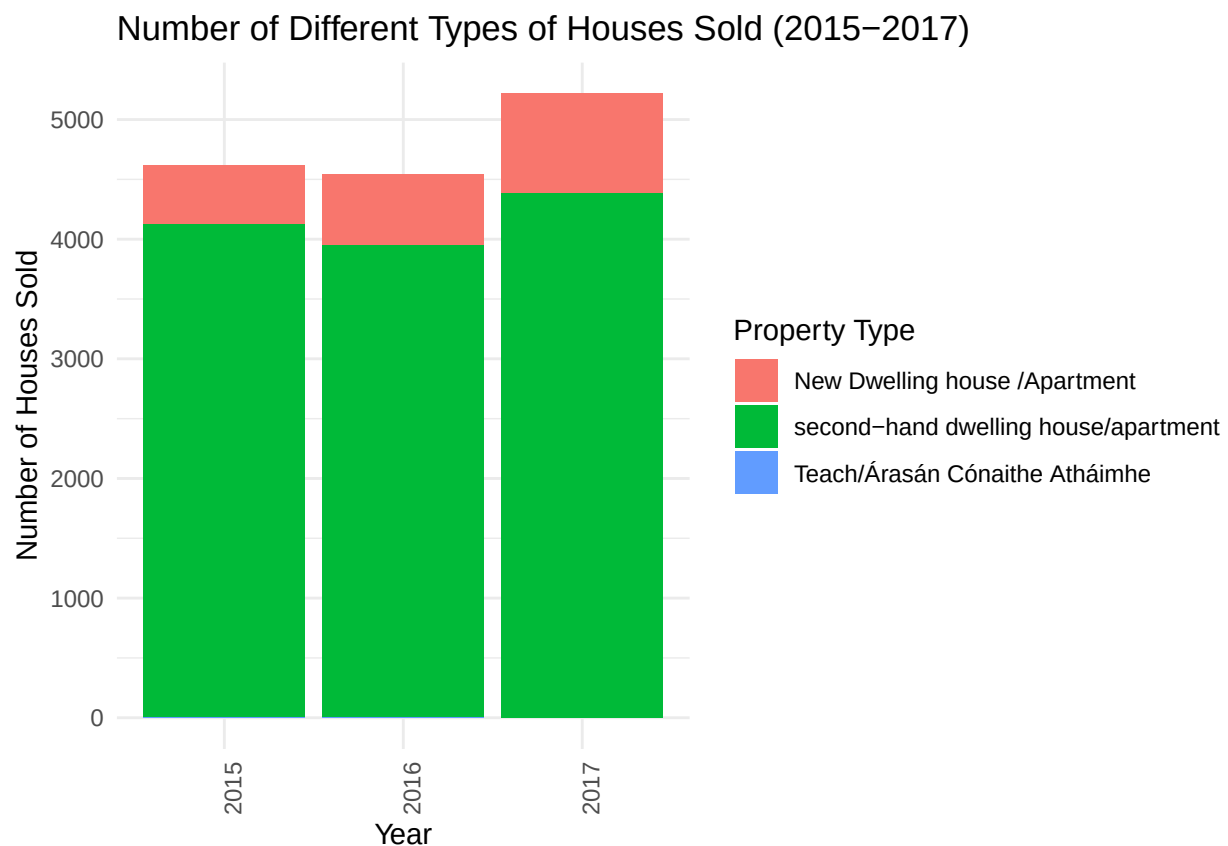


6 Question 1 (e)

```
#Q1.e

# Filter the dataset for sales between 2015 and 2017
sales_2015_2017 <- propertydataset %>%
  filter(year(sale_date) >= 2015 & year(sale_date) <= 2017)

# Create a bar plot showing the number of different types of houses sold by year
ggplot(sales_2015_2017, aes(x = factor(year(sale_date)), fill = property_description)) +
  geom_bar(position = "stack") + # Stack bars for each property type
  labs(
    title = "Number of Different Types of Houses Sold (2015-2017)",
    x = "Year",
    y = "Number of Houses Sold",
    fill = "Property Type" # Legend label
  ) +
  theme_minimal() + # Minimal theme for better aesthetics
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1) # Rotate x-axis labels for readability
  )
```



7 Question 1 (f)

```
#Q1.f

# Remove columns from "num_bedrooms" to "not_full_market_price"
propertydataset <- propertydataset %>%
  select(-num_bedrooms,
         -num_bathrooms,
         -property_size_description,
         -property_description,
         -not_full_market_price)

# Filter rows for Dublin, Cork, Galway, and Semidetached house, then arrange by price
semidetach <- propertydataset %>%
  filter(county %in% c("Dublin", "Cork", "Galway"),
         formatted_description == "Semi-Detached House") %>%
  arrange(price)

# View the first few rows of the filtered and arranged data
head(semidetach)
```

```
##           geohash  sale_date price formatted_address
## 1 gc7rmy3466t5 2015-04-02 27000 13 Foxborough Park, Lucan, Co. Dublin
## 2           2017-07-24 44385 10 Danesfort, Loughrea, Co. Galway
## 3           2016-08-29 48000 12 Egmont Court, Earl Street, Kanturk, Co. Cork
## 4           2018-02-22 55000 52 Mariners Bay, Redbarn, Youghal, Co. Cork
## 5 gce8fzxtczks 2013-09-19 56000 15 Moylaragh Park, Balbriggan, Co. Dublin
## 6 gc1zppp55nje 2012-06-08 57000 21 Dunbar Street, Cork City Centre, Co. Cork
##           raw_address post_code county formatted_description
## 1 13 FOXBOROUGH PARK, LUCAN, DUBLIN Dublin Semi-Detached House
## 2 10 DANESFORT, LOUGHREA, CO GALWAY Galway Semi-Detached House
## 3 12 EGMONT COURT, EARL STREET, KANTURK Cork Semi-Detached House
## 4 52 MARINERS BAY, REDBARN, YOUGHAL Cork Semi-Detached House
## 5 15 Moylaragh Park, Balbriggan, Dublin Dublin Semi-Detached House
## 6 21 Dunbar Street, Cork Cork Semi-Detached House
## vat_exclusive parking parking1 property_size_numeric
## 1 FALSE Has Parking NA
## 2 FALSE Has Parking NA
## 3 FALSE Has Parking NA
## 4 FALSE Has Parking NA
## 5 FALSE No Parking NA
## 6 FALSE Has Parking NA
```

8 Question 1 (g)

```
#Q1.g

# Filter houses greater than €200,000 from Galway, Roscommon, Mayo, or Sligo
houses_more_than_200000 <- propertydataset %>%
  filter(county %in% c("Galway", "Roscommon", "Mayo", "Sligo"),
         price > 200000) %>%
  mutate(year = year(sale_date)) %>%
  filter(!is.na(year)) # Remove rows with missing years

# Summarize by year and county to handle multiple entries per year
houses_summary <- houses_more_than_200000 %>%
  group_by(year, county) %>%
  summarise(avg_price = mean(price, na.rm = TRUE))

# Create a line graph with ggplot
ggplot(houses_summary, aes(x = year, y = avg_price, color = county, group = county)) +
  geom_line() +
  labs(
    title = "Price Trends of Houses Greater than €200,000",
    x = "Year",
    y = "Price (€)",
    color = "County"
  ) +
  theme_minimal() +
  scale_color_manual(values = c(
    "Galway" = "blue", "Roscommon" = "green",
    "Mayo" = "red", "Sligo" = "purple"
  )) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5)
  )
```

Price Trends of Houses Greater than €200,000

